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e passos/dia em comparação com os controles (n=30) (60% e 33% a menos, respectivamente). Além disso, os indivíduos com IC foram ainda mais inativos nos fins de semana em comparação com os dias úteis (AFMV: 6[3-12] vs. 9[4-21] min/dia, $P=0,005$; contagem de passos: 4055 ± 2228 vs. 4550 ± 2366 passos/dia, $P=0,02$). Modelos de regressão indicaram a capacidade funcional como o único determinante do tempo gasto/dia em AFMV ($r^2=0,23$) e um dos determinantes dos passos/dia junto com a idade e o índice de massa corporal ($r^2=0,52$). Os sintomas de depressão e a dispnéia ao esforço foram determinantes do TS/dia ($r^2=0,32$) e do tempo gasto/dia em AF leve ($r^2=0,21$).

Conclusão: Indivíduos com IC são 60% menos ativos que indivíduos saudáveis, sendo ainda mais inativos nos fins de semana. A capacidade funcional foi o principal determinante da variação nos passos/dia e no tempo gasto/dia em AFMV, enquanto a dispnéia ao esforço e os sintomas de depressão explicaram a variação no TS/dia e no tempo gasto/dia em AF leve.

INTRODUCTION

Individuals with heart failure (HF) have been shown to be physically inactive and highly sedentary¹⁻³, a fact apparently influenced by the symptoms of exercise intolerance, fatigue, and dyspnea⁴. However, it is yet unclear what is the magnitude of this reduction in comparison to matched healthy (i.e., non-HF) subjects⁵.

Current literature presents a considerable number of studies investigating the level of physical activity in daily life (PADL) in individuals with HF^{2,3,6,7}. While most of them describe associations, just a few presented comparison with healthy matched peers^{3,8} whereas also only a few compared differences of PADL behavior between week and weekend days⁵. Furthermore, these studies present quite different methodological characteristics such as the type, time of use and body location of the activity monitors (i.e., accelerometers). Accurate information on the PADL level is important since it is known, for example, that a low number of steps per day in this population increases mortality risk.⁹ Moreover, the clinical impact of other variables and outcomes can be further explored, such as the time spent in sedentary behavior and in different intensities of physical activity, as well as the day-to-day variability in these outcomes.

Moreover, looking beyond the PADL profile, the determinants of inactivity and sedentarism in individuals with HF are not yet fully clear. In fact, only a few studies^{10,11} investigated the influence of variables such as age, sex, body mass index, quality of life, presence of ischemic HF and type 2 diabetes mellitus, and therefore the role of these factors in determining the PADL profile has been previously described^{8,10,12,13}. However, the role of many other outcomes such as left ventricle ejection fraction (LVEF), anxiety, depression, dyspnea, fatigue and functional exercise capacity as determinants of the PADL profile has not been studied in depth, so that the extent to which they influence inactivity/sedentarism in this population is still unknown. Therefore, the literature of this field might benefit from further studies which are able to add novel and detailed information on the understanding

population, and in special a deeper understanding of the determinants of their PADL profile. This could be useful to help establishing individualized goals in rehabilitation programs and interventions aiming to promote physical activity (PA) and reduce sedentary time of those individuals.

With more detailed knowledge on these patients' PADL level, interventions aimed at reducing physical inactivity and sedentary behavior can be proposed in a more personalized and assertive way, and rehabilitation programs can generate more clinically relevant and meaningful results to patients.

With that in mind, the aims of this study were: i) to quantify the extent to which individuals with HF are more inactive and sedentary in daily life in comparison to matched controls, both on weekdays and on weekends; and ii) to explore the determinants of variables of inactivity and sedentarism in individuals with HF.

METHODS

The present cross-sectional study was conducted from January to October 2023 at the Laboratory of Research in Respiratory Physiotherapy of the Universidade Estadual de Londrina (UEL), Brazil. A convenience sample comprised individuals with HF attending the cardiology outpatient clinic of the university hospital and healthy adults matched by age, gender and body mass index recruited from the community through social media advertisements. Volunteers were formally contacted by telephone and invited to take part in the research. Following their acceptance, assessments were scheduled. The study obtained approval from the Research Ethics Committee of the institution (No. 6.133.450), and all participants provided consent by signing an informed consent form.

The inclusion criteria for the individuals with HF were as follows: clinical diagnosis of HF (with reduced, preserved or intermediate LVEF), 18 years or older, clinical stability in the last three months (i. e., no hospitalization in the previous three months and current pharmacological optimization, as gathered in the medical records), absence

s included failure to complete
l for any reason (e.g., allergy to
onitor strap, expressed desire to
y).

All participants had their PADL measured using an activity monitor (Actigraph wGT3X, Actigraph, USA), which they were instructed to wear above their right hip (attached to a strap) during waking hours (except during water activities) for seven consecutive days. Data were recorded in 1-minute epochs. Data processing involved several steps on the per-minute accelerometer records. First, non-wear time, defined as 60 minutes or more of consecutive zeroes in activity counts, was excluded for each recorded day. Records were also removed if data reliability was flagged in the original dataset, including extreme activity count values. The total number of valid minutes for each recorded day was assessed, and only days with at least 8 valid hours were included. A minimum of four valid days was required for analysis.

PADL variables were step count (steps/day), time spent/day (in minutes) in sedentary postures (i.e., sitting, reclining and lying positions) and time spent/day (in minutes) at different intensities based on published intensity thresholds:^{14,15} sedentary activities (≤ 1.5 metabolic equivalents of task [MET]), light PA (1.6 to 2.9 MET) and moderate-to-vigorous PA (≥ 3 MET [MVPA]). In addition to showing sedentary time in minutes/day, it was also calculated as percentage of the total assessment time during the day. All variables were averaged over all valid assessment days. Further, total MVPA was described as the sum of MVPA in all valid days (min/week). The averages of weekdays and weekends were separately considered for subanalysis.

Functional exercise capacity was assessed using the six-minute walking test (6MWT) according to international guidelines¹⁶. Reference values for the Brazilian population were used¹⁷. The Borg scale was used to assess dyspnea and fatigue during the test. In order to assess the health-related quality of life in individuals with HF, the Brazilian version of the Minnesota Living with Heart Failure Questionnaire (MLHFQ) was used¹⁸.

The presence of comorbidities was assessed using a list developed by the authors, relying on both Charlson and Elixhauser indices^{19,20}, as well as other comorbidities more prevalent among patients with HF²¹. They were self-reported and verified in medical records. Additional clinical information such as medication, the New York Heart Association (NYHA) functional classification and LVEF were also recorded. Finally, other assessments included anthropometric (body mass index, BMI) and socio-demographic aspects (sex, age, work status, marital status), and anxiety and depression symptoms (Hospital Anxiety and Depression scale, HADS)²².

Normality in data distribution was checked with the Shapiro-Wilk test. Continuous variables were described as absolute and relative frequency (%), mean $\pm$ standard

Pearson or Spearman correlation coefficients, according to the normality in data distribution. Variables with a correlation coefficient (r) of at least, 0.30 were included in multiple linear regression models to predict each variable of PADL (dependent variables). Statistical significance was set at $P \leq 0.05$. All statistical analyses were performed by the SPSS® Statistics 22.0 software (IBM Corporation, USA).

Power was calculated *post hoc* using the software G*Power 3.1.9.7 (Heinrich-Heine University Duesseldorf, Germany) with a t test comparison between two independent means for the variable step count (steps/day), since it is a broad reflection of PADL of different intensities.

RESULTS

In total, 76 individuals were included in the study (46 with HF and 30 controls). Two outliers among individuals with HF were identified in the PADL analyses and excluded; hence, 44 individuals with HF were analyzed. The power *post hoc* analysis for the outcome step count ($\alpha=0.05$ and effect size=0.80) showed that the studied sample had a power of 0.95 (95%).

Characteristics of the studied individuals, as well as the intergroup comparisons, are described in Table 1. As expected, individuals with HF presented more comorbidities and worse functional exercise capacity than controls.

Concerning the PADL profile, individuals with HF were considerably less physically active than healthy controls (Table 2). They spent less than half of the time in MVPA in comparison to the control group, considering both time/day and total time/week (60 and 70% less, respectively). Similarly, they presented one third less steps/day than controls. On the other hand, there were no statistically significant differences between individuals with HF and healthy controls in terms of sedentary time/day and time spent/day in light PA.

Further analyses showed that individuals with HF were even less active during weekends in comparison to weekdays (Figure 1), as evidenced by the time spent/day in MVPA (6 [3 – 14] vs. 10 [4 – 22] min/day, respectively; $P=0.014$) and step count (4187 $\pm$ 2368 vs. 4615 $\pm$ 2377 steps/day; $P=0.060$). Furthermore, there were no significant differences between weekends and weekdays in time spent/day in light PA (252 $\pm$ 115 vs. 260 $\pm$ 99 min/day; $P=0.477$) and sedentary time (498 $\pm$ 93 vs. 509 $\pm$ 92 min/day; $P=0.329$).

Table 3 shows that the distance covered in the 6MWT, exertional dyspnea and fatigue, age, NYHA class, MLHFQ and HADS scores correlated weakly to moderately (and significantly) with different PADL variables in individuals with HF. Conversely, sex and LVEF did not correlate significantly with any of the PADL variables. Sedentary time in minutes/day only correlated significantly with the Borg dyspnea scale at the end of the 6MWT, although only weakly ($r=0.31$; $P=0.044$). No significant correlation of sedentary time in minutes/day with any other variable

		65±11	62±10	0.200
		21(48%)/23(52%)	11 (37%)/19(63%)	0.160
	n (%)	28 (64%)	20(67%)	0.197
Retired, n (%)		32 (72%)	8 (27%)	0.003
Clinical profile				
NYHA class, n (%)				
I/II		36 (82%)	N/A	-
III		8 (18%)	N/A	-
LVEF, %		47±15	N/A	-
BMI, Kg.m ⁻²		29.6±5.3	28.7±5.1	0.457
Comorbidities, n (%)				
Hypertension		25 (83%)	11 (37%)	0.0001
Diabetes		17 (57%)	4 (13%)	0.050
Obesity		13 (43%)	10 (33%)	0.165
Depression		19 (63%)	5 (17%)	0.173
Smoker / ex-smoker		20 (67%)	7 (23%)	0.0001
Medication, n (%)				
Diuretic		23 (53%)	N/A	-
Beta blocker		20 (47%)	N/A	-
ACEI/ARB		21 (49%)	N/A	-
ARNi		10 (23%)	N/A	-
Symptoms				
HADS anxiety, points		5 [2-10]	5 [3-6]	0.908
HADS depression, points		5 [2-7]	4 [2-7]	0.674
Functional capacity				
6MWT, % pred		80±16	104±11	0.001
Health-related quality of life				
MLHFQ total, points		42 [23-62]	N/A	-
MLHFQ-physical, points		21 [5-31]	N/A	-
MLHFQ-emotional, points		8 [3-15]	N/A	-

Results are shown as mean ± standard deviation, median [interquartile range] or absolute number (% of the sample). NYHA: New York Heart Association; LVEF: left ventricular ejection fraction; BMI: body mass index; ACEI: angiotensin-converting enzyme inhibitors; ARB: angiotensin II receptor blocker; ARNi: angiotensin receptor-neprilysin inhibitor; HADS: Hospital Anxiety and Depression Scale; 6MWT: six-minute walking test; MLHFQ: Minnesota Living with Heart Failure Questionnaire.

Table 2 – Variables of physical activity in daily life in individuals with heart failure and healthy matched controls.

	Heart Failure (n=44)	Controls (n=30)	<i>P</i> value HF vs controls	Δ control - HF*
Wearing time, min/day	774±97	804±79	0.165	30 (-13 – 72)
Sedentary Time, min/day	510±81	490±94	0.336	-15 (-57 – 28)
Sedentary Time, % day	66±11	61±12	0.076	-4 (-1 – 1)
Step Count, steps/day	4404±2259	6594±3163	0.001	2109 (845 – 3374)
Light PA, min/day	256±101	294±98	0.119	37 (-10 – 84)
MVPA, min/day	9 [4-17]	22 [7-31]	0.008	13 (2 – 15)
Total MVPA, min/week	41 [16-104]	118 [48-181]	0.001	82 (16 – 99)

Results are shown as mean ± standard deviation or median (interquartile range). * Δ control - HF represents the median (95% confidence

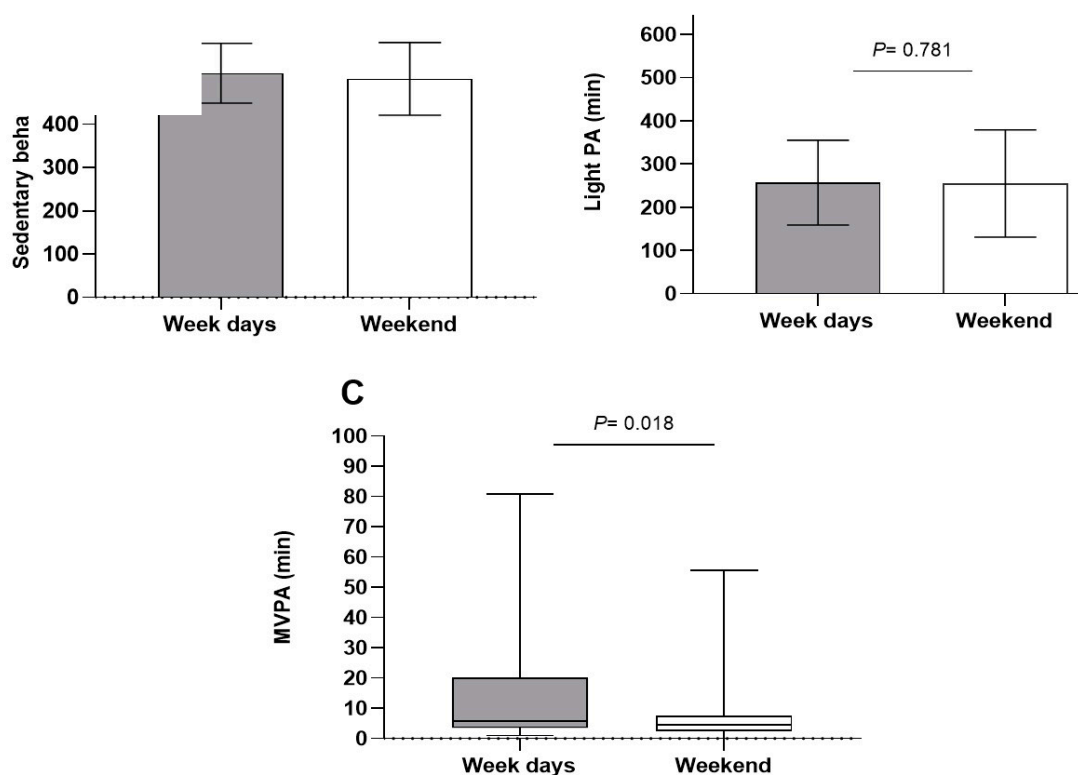


Figure 1 – Comparison of the PADL variables between weekdays and weekends in patients with HF. A) sedentary time; B) light physical activity; C) moderate-vigorous physical activity.

Table 3 – Correlations between the variables of physical activity in daily life and other outcomes in individuals with heart failure.

	Sedentary Time (% day)	Step Count (steps/day)	Light PA (min/day)	MVPA (min/day)
Age, yrs	0.41*	-0.49*	-0.30*	-0.51*
Sex	0.11	0.20	-0.10	0.28
BMI, Kg.m ⁻²	0.13	-0.36	-0.12	-0.17
LVEF, %	0.20	-0.17	-0.24	-0.03
NYHA class	0.28	-0.33*	-0.20	-0.19
MLHFQ total, pts	0.33*	-0.38*	-0.25	-0.23
HADS Anxiety, pts	0.31*	-0.22	-0.25	-0.08
HADS Depression, pts	0.40*	-0.26	-0.33*	-0.16
6MWT, %pred	-0.38*	0.50*	0.33*	0.51*
Dyspnea_6MWT, pts*	0.39*	-0.48*	-0.31*	-0.43*
Fatigue_6MWT, pts*	0.17	-0.32*	-0.07	-0.32*

correlations shown in Table 3 were performed with the sedentary time in % of the day, a variable that was better

After identifying significant correlations, Table 4 shows four multiple regression models developed to ascertain the

		112,226.373 (4,696.857 - 17,482.890)	-	0.001
		17.548 (15.274 - 86.208)	0.352	0.006
BMI, Kg.m ⁻²		47.753 (-277.302 - -84.276)	-0.419	0.001
Age, yrs		26.671 (-138.114 - -30.307)	-0.386	0.003
Dependent variable: MVPA, min/day	0.23			
Constant		-9.415 (-21.529 - 2.699)	-	0.124
6MWT, % pred		0.252 (0.104 - 0.400)	0.478	0.001
Dependent variable: Light PA, min/day	0.21			
Constant		374.944	-	< 0.0001
HADS Depression, pts		-7.982 (-14.842 - -1.122)	-0.328	0.024
Dyspnea_6MWT#, pts		-7.652 (-15.037 - -0.267)	-0.292	0.043
Dependent variable: Sedentary Time, % day	0.32			
Constant		0.488	-	<0.0001
Dyspnea_6MWT#, pts		0.011 (0.004 - 0.019)	0.385	0.005
HADS Depression, pts		0.010 (0.003 - 0.018)	0.380	0.005

count and time spent/day in MVPA and light PA, as well as sedentary time (in % of the day, due to the abovementioned lack of relevant correlations of sedentary time in minutes/day with the other studied variables). Symptoms of depression and exertional dyspnea explained 32% of the variation in sedentary time ($R^2 = 0.32$; $P < 0.0001$), as well as 21% of the variation of light PA ($R^2 = 0.21$; $P < 0.0009$). Age, BMI and 6MWT explained 52% of the variation in step count ($R^2 = 0.52$; $P < 0.001$), whereas only the 6MWT was a significant predictor of MVPA, accounting for 23% of its variation ($R^2 = 0.23$; $P < 0.001$).

DISCUSSION

The present study showed that, in comparison to healthy marched controls, the PADL profile of individuals with HF is characterized by marked physical inactivity, especially in terms of time spent in MVPA (which was reduced by 60% - 70%) and steps/day. On the other hand, differences between individuals with HF and controls were much less pronounced in terms of sedentary behavior and light PA. Further, in individuals with HF, there is even more marked inactivity in weekend days than in weekdays. Symptoms of depression and exertional dyspnea were determinants of time spent in sedentary behavior and light PA. Functional exercise capacity (6MWT) was solely a main determinant of time spent in MVPA, whereas 6MWT, age and BMI were determinants of step count in these patients.

Although physical inactivity has been previously described in individuals with HF, the magnitude of this inactivity in comparison to healthy controls has not been

results corroborate with the present research^{2,3}, whereas other studies presented different results^{3,6,8}. In face of the challenge of physical inactivity, a possible way to improve the MVPA in these individuals is through exercise training, especially if performed according to guidelines (i.e., three times a week, at least 30 minutes of aerobic exercise)^{4,23}.

In general, individuals with HF in the present study did not reach the goal of steps/day recommended for people living with chronic illness in order to obtain health benefits, i.e., 6500 steps/day²⁴. They also did not reach the mean of 5040 steps/day as in the study by Jordan et al.⁷, although the present sample has a higher number of steps/day than other previous studies⁸. The number of steps/day is commonly linked with MVPA, although previous research²⁵ has suggested that its link with light PA is more likely to yield achievable goals in terms of improvement of PA level, perhaps impacting on mortality risk.

The profile of PADL of individuals with HF in this study was heterogenous throughout the week. Time spent in MVPA and number of steps/day were lower in weekends days. This result could be likely explained by the fact that some individuals with HF still had to work during weekdays, so in the weekends they may generally prefer lower motion activities in their free time to rest, as previously suggested in patients with Chronic Obstructive Pulmonary Disease²⁶. On the other hand, the opposite may happen in healthy adults²⁷, in whom the free time on the weekends may be linked with activities of higher intensities. Schwendinger et al.⁵ also found this pattern of more marked inactivity on weekend days in individuals with HF, although sedentary behavior and light PA were

e groups HF and control. A mean hours/day in sedentarism was a result similar to other studies with HF³, although results up to an average of 12 hours/day in sedentary time for adults has been described²⁸. This similar behavior among the groups raises questions about the fact that sedentary behavior may be more a choice than a consequence of HF. A similar pattern can be seen in individuals with lung diseases such as bronchiectasis and severe asthma²⁹.

The present results on the determinants of PADL variables in HF are a new way to look at inactivity and sedentary behavior in this population. Table 4 shows that 21% of the variation of time spent in light PA and 32% of the variation of time spent in sedentary behavior were explained by symptoms of depression and dyspnea during exertion. Hence, light PA and sedentary behavior were closely connected to the functional limitation of these patients by dyspnea, whereas depression was already described as a limitation for motivation and readiness to PA in individuals with HF³⁰. Conversely, number of steps/day and especially time spent in MVPA were significantly predicted by exercise capacity (6MWT), corroborating the concept that the performance of MVPA in daily life in individuals with chronic cardiorespiratory problems is strongly linked to better physical conditioning. In this sense, exercise training may be required in many patients in order to impact on improving MVPA, whereas symptom management (i.e., dyspnea and depression) may be one of the keys to reverse sedentary behavior.

Potential limitations of the present study are the small size and convenience nature of the sample, which may reduce its representativity. In addition to the single-center design in a country with considerable socioeconomic discrepancies, these factors can hinder the generalizability of the results. Further, the relatively preserved functional exercise capacity of the sample may also differ from previous reports, despite the fact that the majority had not been involved in cardiac rehabilitation programs. Finally, outcomes may differ among different types of activity monitor, and results from this study may not be applicable for comparison with other monitors from previous and future reports. These factors may eventually limit the external validity of the present findings.

CONCLUSION

Individuals with HF are 60%-70% less active than matched controls, and even more markedly on weekends. Functional exercise capacity was the main determinant of variation in MVPA and steps/day, whereas dyspnea and depression helped explain variation in ST/day and light PA. Future studies may take these results into account in order to finetune strategies to change these patients' profile of inactivity. For example, strategies to reduce physical

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Individual contribution of the authors:

Study conception and design: KD, NH

Data collection: KD

Data analysis and interpretation: KD, NH

Manuscript writing: KD, LB

Critical review of the text: NH, FP

Final approval of the manuscript*: KD, LB, FP, NH

Statistical analysis: KD, NH

Overall responsibility for the study: KD, FP, NH

*All authors have read and approved the final version of the
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